CRM Sales Management System

Pure Hibernate JPA - Console Based Application

1. Core Functionalities (Step-by-Step Design)

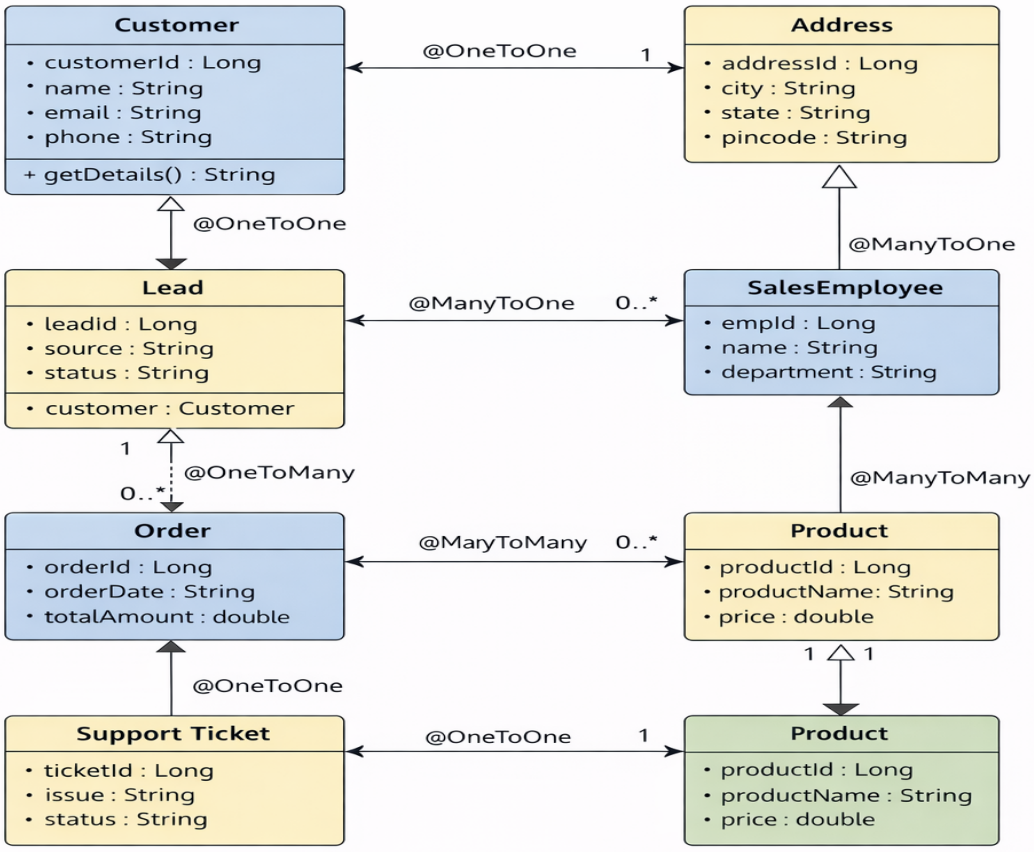
The CRM system must handle customer management, lead tracking, order processing, product management, and support ticket handling. Each functionality must follow proper entity creation, relationship mapping, transaction handling, and service method design.

- Register Customer → Class: CustomerService → Method: registerCustomer(String name, String email, String phone)
- Add Address → Class: CustomerService → Method: addAddressToCustomer(Long customerId, Address address)
- Create Lead → Class: LeadService → Method: createLead(String name, String source, String contactInfo)
- Assign Lead → Class: LeadService → Method: assignLeadToEmployee(Long leadId, Long employeeId)
- Convert Lead → Class: LeadService → Method: convertLeadToCustomer(Long leadId)
- Add Product → Class: ProductService → Method: addProduct(String name, double price)
- Place Order → Class: OrderService → Method: placeOrder(Long customerId, List productIds)
- Raise Support Ticket → Class: TicketService → Method: raiseTicket(Long orderId, String issueDescription)
- View Employee Performance → Class: ReportService → Method: getEmployeePerformance(Long employeeId)

2. Entity Class Design

Each entity must be placed inside the 'entity' package and annotated with @Entity. Primary keys must use @Id and @GeneratedValue. Relationships must use proper mapping annotations.

- Customer → Fields: id, name, email, phone, Address address, List orders
- Address → Fields: id, street, city, state, zipCode
- Lead → Fields: id, name, source, contactInfo, SalesEmployee employee
- SalesEmployee → Fields: id, name, department, List leads
- Product → Fields: id, name, price
- Order → Fields: id, orderDate, totalAmount, Customer customer, List products
- SupportTicket → Fields: id, issueDescription, Order order



3. Service Layer Design

Service classes must contain business logic and transaction handling. Each method must begin a transaction, perform persistence operation, commit transaction, and handle rollback in case of exception.

- CustomerService → Handles customer registration, update, delete
- LeadService → Handles lead creation, assignment, conversion
- OrderService → Handles order placement and total calculation
- ProductService → Handles product creation
- TicketService → Handles support ticket operations
- ReportService → Handles JPQL queries for analytics

4. Main Class Design (Console Flow)

Main class name: CRMApplication Steps inside main(): 1. Create EntityManagerFactory 2. Create EntityManager 3. Create Scanner for user input 4. Display menu inside while loop 5. Use switch-case for each option 6. Call appropriate service method 7. Close EntityManager and EntityManagerFactory before exit

- 1 → Call customerService.registerCustomer()
- 2 → Call customerService.addAddressToCustomer()
- 3 → Call leadService.createLead()
- 4 → Call leadService.assignLeadToEmployee()
- 5 → Call leadService.convertLeadToCustomer()
- 6 → Call productService.addProduct()
- 7 → Call orderService.placeOrder()
- 8 → Call ticketService.raiseTicket()
- 9 → Call reportService.getEmployeePerformance()
- 10 → Exit application